

Ujjwal Varshney

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SUMMARY

Backend & Full-Stack Developer with 1+ year of experience building enterprise-grade distributed systems for the oil & gas industry. Expertise in ASP.NET Core, event-driven microservices, and RESTful/GraphQL APIs. Hands-on experience with AI/RAG integration, OAuth 2.0 security, and multi-tenant architecture serving 10+ enterprise clients with 99.9% high availability.

EXPERIENCE

Software Developer

Petro IT

Jan. 2025 – Present

Gurugram, India

- Architected multi-tenant reporting platform in ASP.NET Core with dynamic database routing and API Gateway pattern, serving 10+ enterprise clients on a single deployment with full request-scoped isolation.
- Built event-driven microservices pipeline with reflection-based handler discovery, parallel/sequential execution, and message queue processing (PostgreSQL) across 15+ entity handlers — replacing MediatR.
- Designed Redis Sentinel caching with Lua atomic scripts, MongoDB fallback, and load balancing support — achieving 99.9% high availability with zero application crashes.
- Developed 22+ report services using complex MongoDB aggregations, GraphQL and REST endpoints, connection pooling (500 max), and concurrent processing via Task.WhenAll(); documented all endpoints with Swagger/OpenAPI.
- Implemented OAuth 2.0 and JWT middleware with semaphore-protected token refresh, CSP/security headers, MFA (TOTP), and AWS S3 integration for secure document storage.
- Wrote Unit Testing and Integration Testing suites (xUnit, NUnit) covering auth flows, report services, and cache fallback logic.
- Built AI Database Chatbot using LLM/RAG with NL-to-SQL translation, Apache Kafka event streaming for async query processing, supporting 35+ languages and 90%+ query accuracy across 100+ tables.

PROJECTS

Enterprise Reporting Platform (Oil & Gas) | .NET 8 · MongoDB · Redis · PostgreSQL · Docker

2025

- Multi-tenant 6-project .NET solution with API Gateway pattern, request-scoped DB routing, CI/CD via GitHub Actions, and PuppeteerSharp PDF generation supporting 100k+ row Excel exports.
- Dual-layer caching (Redis + MongoDB fallback) with high availability design; 22+ report services with Swagger-documented APIs and i18n for 15+ languages.

Enterprise UI Framework (ERP) | Angular 8 · TypeScript · RequireJS · Express.js · OAuth 2.0

2025

- Hybrid Angular/AMD SPA with 45+ modules, drag-and-drop form builder, workflow state machine, and virtual scrolling for 100k+ rows; OAuth 2.0 and MFA (TOTP, device tracking) integrated.
- BI dashboard via Apache Superset SDK, load balanced deployment via Kubernetes CDK8s, and multi-language support for 15+ languages.

Conversational AI Database Assistant | Python · FastAPI · LLaMA 3 · RAG · Ollama · spaCy · RabbitMQ

2025

- NL-to-SQL engine with RAG architecture, dynamic schema introspection, auto JOIN optimization, and RabbitMQ-based async query processing — 90%+ accuracy across 100+ tables.
- NLP pipeline using spaCy NER, FAISS vector embeddings, few-shot prompting, and session-persistent conversation memory with Swagger-documented REST API.

TECHNICAL SKILLS

Languages: C#, Python, TypeScript, JavaScript, Java, SQL

Backend: ASP.NET Core, .NET 8, FastAPI, Node.js, Express.js, RESTful APIs, GraphQL, API Gateway, Swagger/OpenAPI

Frontend: Angular, React, jQuery, RequireJS/AMD, HTML5, CSS3

Databases: SQL Server, PostgreSQL, MongoDB, Redis, Elasticsearch, Query Optimization, Load Balancing

Messaging: Event-Driven Architecture, Message Queues (PostgreSQL), RabbitMQ, Apache Kafka

Auth: JWT, OAuth 2.0, MFA (TOTP, Digital Signature), Security Headers (CSP, X-Frame-Options)

Testing: Unit Testing (xUnit, NUnit), Integration Testing, Postman, API Testing

AI / ML: LLM, RAG, LangChain, FAISS, ChromaDB, Ollama, spaCy, Vector Databases, Prompt Engineering

DevOps: Docker, Kubernetes, Azure DevOps, GitHub Actions, Git, CI/CD, Grafana, Prometheus

Other: System Design, Multi-tenant Architecture, High Availability, Agile/Scrum, Code Reviews

EDUCATION

GLA University

B.Tech in Computer Science & Engineering — CGPA: 8.32 / 10

Mathura, India

Sep. 2021 – Jul. 2025

ACHIEVEMENTS

- Solved 350+ DSA problems on LeetCode, HackerRank, and HackerBlocks.
- 5-star in Java on HackerRank (Top 5%); 3-star in C and Python.
- Selected for Amazon ML Summer School 2024 — training in ML, Deep Learning, and NLP.
- Chegg Subject Matter Expert — resolved 200+ CS queries with 4.5+ rating.